Control Systems

(Automatic Control, Linear Control, Engineering Control)

Basic assumption → LTI systems

We have a

Real system (Reality) → Model (approximation) Linear D.E → Controller Design And Aulysiz

(Representation and simplification methods)

Controller Design And Aulysiz:
- Time domain
- Root locus meth.
- ↳ Freq domain meth.
- Frequency domain
- a loop

Output should be

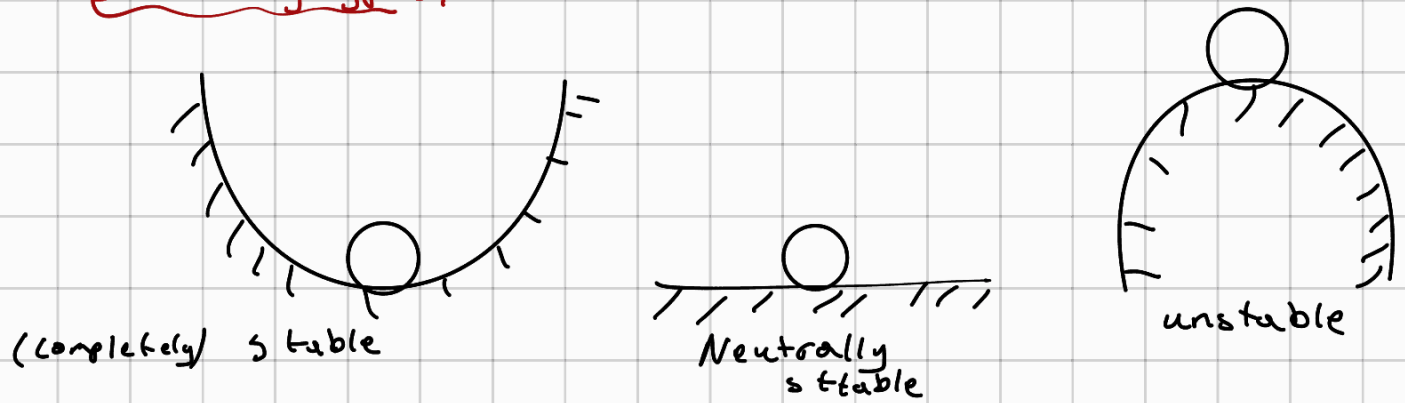
1- Accurate (error) ← Partical solution

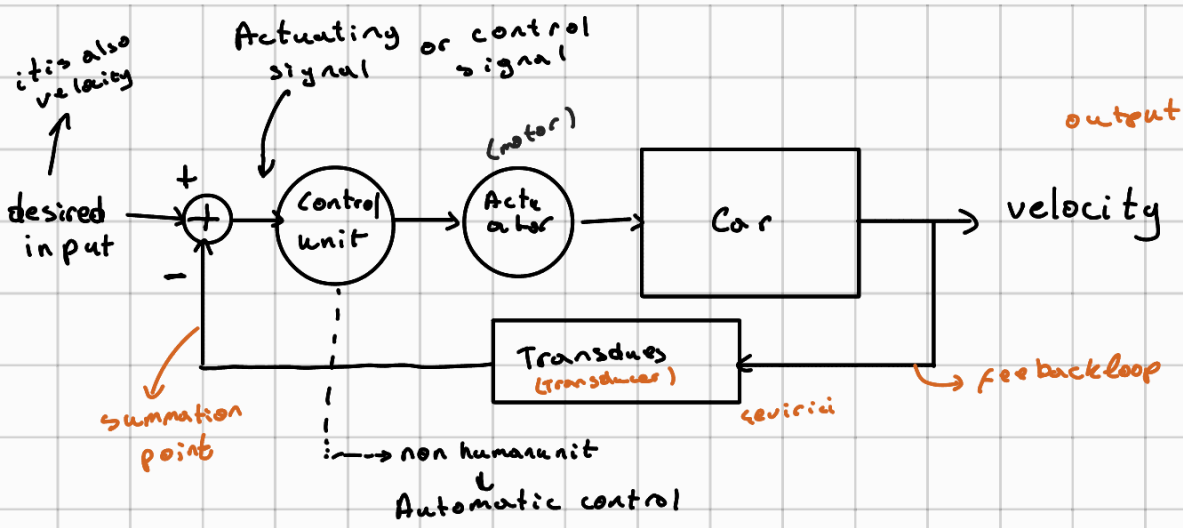
2- Quality (of the output) (%O.S, t_s, t_p) ← Homogen solution

* 3- Stability

To control asystem: it should be dynamic, stable,

Stability types:





Classification for control systems

- Architecture → open loop
  - ↳ closed loop : feed back
- Energy Source in the System → Active (always need energy source)
  - ↳ passive (you dont need energy sorce)
- Input-output → SISO (Single input single-output) (classial control)
  - ↳ SIMO (" " multiple ")
  - ↳ MISO
  - ↳ MIMO

(classial control) → Root locus, Bode-plot (Freq)

Modern control (time domain) → state-space

SISO example : Regulators, Amplifers, thermositor, dc motors water controller in a pool,

MIMO Example : Autonom vehicle, Electrial distrubition system, aircraft systems, Internet Router, Camera, Temperature System of (Bma)

SIMO Example : Radar systems, Powerbank

MISO Example : A computer temp sytem with multiple sensors and one fan